

Control System: Phases 0 and 1

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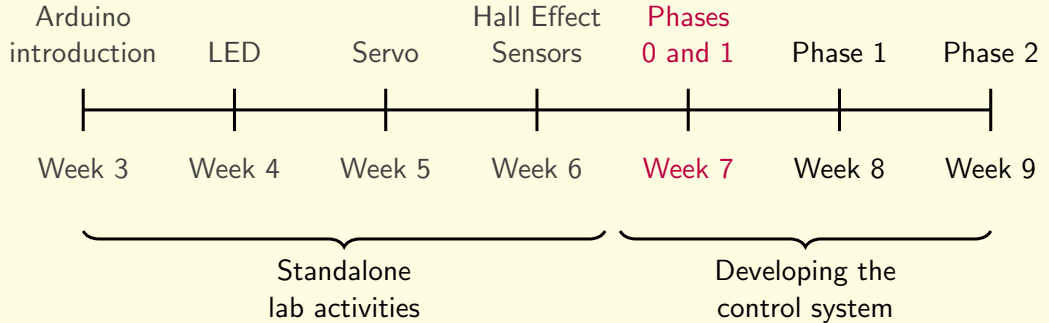
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MAC 233: **11/11/2026**



University of
Sheffield

MAC 233: LABS OVERVIEW

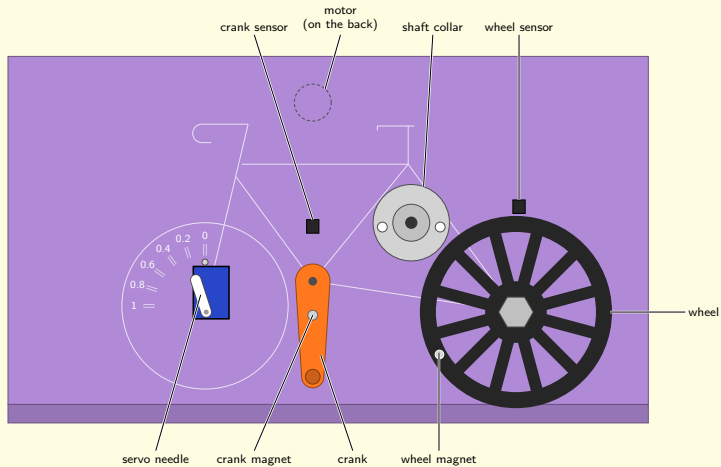


THE NEXT THREE WEEKS

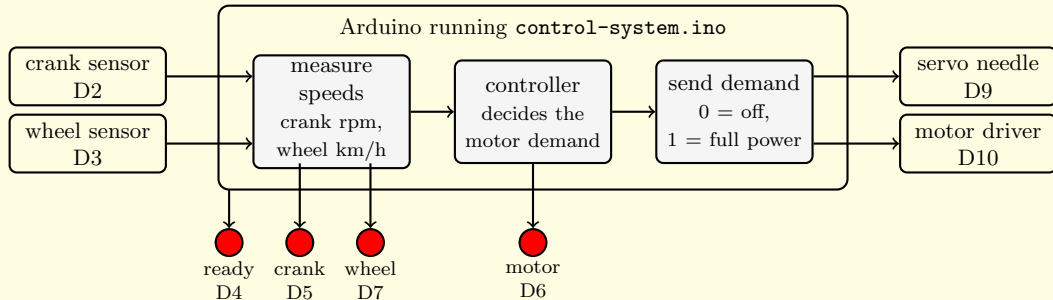
Week	Phase	Mini rig?
7	0: how the control system works	No
	1: build steps 1–3	No, use a magnet
8	1: build steps 4–5	Only for step 5
9	2: developing the control system	Yes

Your rig isn't finished? That's fine: steps 1–4 only need your Arduino kit and a magnet.

THE MINI RIG



THE CONTROL SYSTEM

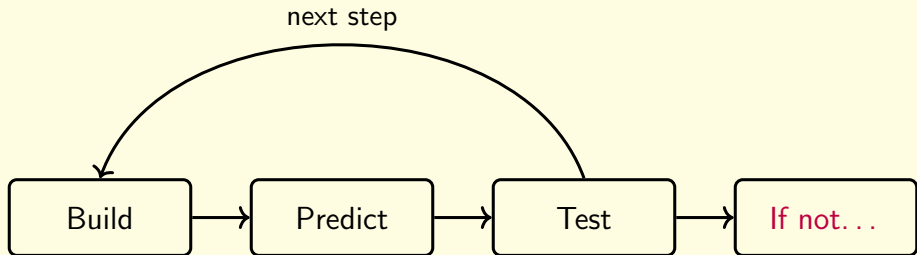


Only help while the rider is pedalling, and never above 25 km/h.

BUILDING IT: ONE PART AT A TIME

Build everything, then test it: if it doesn't work, where is the fault?

Build one part, then test it: the fault is in what you just added.

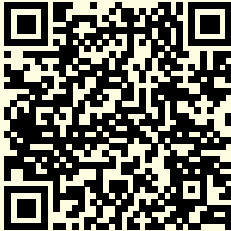


Phase 0: read through the sketch, then answer the check questions with a partner.

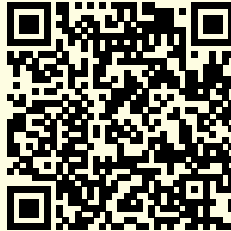
Phase 1: build steps 1–3 on the bench. Pass the magnet by hand to test the sensors.

Upload the sketch once, at the start of step 1. You won't change it until Phase 2.

Worksheet



Sketch



Nano Every pinout



Problems? Raise your hand and someone will come and help you.